INCH-POUND
MIL-DTL-24643/3F
27 November 2023
SUPERSEDING
MIL-DTL-24643/3E
1 October 2009

DETAIL SPECIFICATION SHEET

CABLE, ELECTRICAL, -4 DEGREES FAHRENHEIT (-20 DEGREES CELSIUS) TO +194 DEGREES FAHRENHEIT (+90 DEGREES CELSIUS), 600 VOLTS, TYPES LSSHOF, LSDHOF, LSTHOF, AND LSFHOF

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and MIL-DTL-24643.

Construction (Non-Watertight)

First - Copper conductor, coated or uncoated (see table I for size).

Second - Separator may be used at manufacturer's option where required to provide free-stripping insulation.

Third - Thermoset insulation (see table I for wall thickness). Standard identification code applied by method 1 or method 3. If identification method 1 is used, insulation color shall be black.

NOTE: A black colored tape, providing 100 percent coverage, may be used on type LSSHOF cables, size 23 and larger, in lieu of black insulation.

Fourth - The specified number of conductors (see table I) shall be cabled together with a left-hand lay as required for the completed cable to meet the twisting and bending endurance tests. On multi-conductor types, fillers may be used as necessary to form a firm, well-rounded assembly; the cabling sequence shall be consecutive.

NOTE: For type LSSHOF cables, omit fourth step.

Fifth - An optional binder.

Sixth - Cross-linked polyolefin jacket. See table I for thickness.

NOTE: Manufacturer's identification tape is required (see 3.4.9 of MIL-DTL-24643). For LSSHOF cable, the identification tape information must be accessible and legible.

Distribution Statement A. Approved for public release: distribution is unlimited.

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TABLE I. Details.

Military part no. M24643/3	Type and size	Conductor size		Insulation thickness min. avg. (inch)	Number of conductors	Jacket wall thickness min. avg. (inch)	Overall diameter (inch)		Roller diameter for bending endurance max. (inch)	Conductor resistance (ohms)	Insulation resistance (megohms)
		Navy STD	AWG/MCM				min.	max.			
-01UN	LSSHOF-3	--	16 (class M)	0.031	1	0.030	0.195	0.210	N/A	4.28	300
-02UN	LSSHOF-23	--	7 (class K)	0.040	1	0.040	0.440	0.460	N/A	0.546	300
-03UN	LSSHOF-60	60 (304)	--	0.050	1	0.050	0.570	0.600	N/A	0.187	250
-04UN	LSSHOF-150	150 (760)	--	0.070	1	0.050	0.830	0.870	N/A	0.0747	200
-05UN	LSSHOF-200	200 (988)	--	0.070	1	0.050	0.940	0.980	N/A	0.0575	200
-06UN	LSSHOF-250	250 (1254)	--	0.070	1	0.070	1.035	1.085	N/A	0.0453	150
-07UN	LSSHOF-500	--	500 MCM (class K)	0.090	1	0.070	1.380	1.450	N/A	0.0234	100
-08UN	LSSHOF-650	--	650 MCM (class G)	0.100	1	0.070	1.540	1.610	N/A	0.0174	100
-09UN	LSSHOF-800	800 (4033)	--	0.100	1	0.070	1.600	1.670	N/A	0.0141	100
-10UN	LSDHOF-3	--	16 (class K)	0.031	2	0.040	0.405	0.425	1.3	4.36	300
-11UN	LSDHOF-4	--	14 (class K)	0.031	2	0.040	0.440	0.460	1.4	2.73	250
-12UN	LSDHOF-6	--	12 (class K)	0.031	2	0.040	0.490	0.510	1.5	1.72	200
-13UN	LSDHOF-9	9 (90)	--	0.031	2	0.050	0.540	0.570	1.7	1.24	200

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TABLE I. Details – Continued.

Military part no. M24643/3	Type and size	Conductor size		Insulation thickness min. avg. (inch)	Number of conductors	Jacket wall thickness min. avg. (inch)	Overall diameter (inch)		Roller dia. for bending endurance max. (inch)	Conductor resistance (ohms)	Insulation resistance (megohms)
		Navy STD	AWG/MCM				min.	max.			
-14UN	LSDHOF-14	14 (140)	--	0.040	2	0.050	0.675	0.705	2.1	0.802	200
-15UN	LSDHOF-23	--	7 (class K)	0.040	2	0.050	0.820	0.860	3.0	0.560	150
-16UN	LSDHOF-30	--	5 (class K)	0.050	2	0.050	0.920	0.960	3.5	0.351	150
-17UN	LSDHOF-83	83 (418)	--	0.070	2	0.070	1.390	1.450	7.0	0.139	150
-18UN	LSDHOF-250	250 (1254)	--	0.070	2	0.100	2.000	2.100	N/A	0.0462	100
-19UN	LSDHOF-400	400 (2052)	--	0.090	2	0.100	2.400	2.500	N/A	0.0283	100
-20UN	LSTHOF-3	--	16 (class K)	0.031	3	0.040	0.430	0.450	1.4	4.36	300
-21UN	LSTHOF-4	--	14 (class K)	0.031	3	0.040	0.460	0.480	1.4	2.73	250
-22UN	LSTHOF-6	--	12 (class K)	0.031	3	0.050	0.520	0.550	1.7	1.72	200
-23UN	LSTHOF-9	9(90)	--	0.031	3	0.050	0.570	0.600	1.8	1.24	200
-24UN	LSTHOF-14	14 (140)	--	0.040	3	0.050	0.720	0.750	2.3	0.802	200
-25UN	LSTHOF-23	--	7 (class K)	0.040	3	0.050	0.860	0.900	3.5	0.560	150
-26UN	LSTHOF-42	42 (209)	--	0.070	3	0.070	1.200	1.250	4.5	0.277	150
-27UN	LSTHOF-150	150 (760)	--	0.070	3	0.100	1.740	1.820	N/A	0.0762	125
-28UN	LSTHOF-250	250 (1254)	--	0.070	3	0.100	2.140	2.240	N/A	0.0462	100
-29UN	LSTHOF-400	400 (2052)	--	0.090	3	0.100	2.680	2.800	N/A	0.0283	100

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TABLE I. Cable details – Continued.

Military part no. M24643/3	Type and size	Conductor size		Insulation thickness min. avg. (inch)	Number of conductors	Jacket wall thickness min. avg. (inch)	Overall diameter (inch)		Roller dia. for bending endurance max. (inch)	Conductor resistance (ohms)	Insulation resistance (megohms)
		Navy STD	AWG/MCM				min.	max.			
-30UN	LSTHOF-500	--	500 MCM (class K)	0.090	3	0.100	2.920	3.100	N/A	0.0230	100
-31UN	LSTHOF-600	--	600 MCM (class G)	0.100	3	0.100	2.980	3.150	N/A	0.0200	100
-32UN	LSFHOF-3	--	16 (class K)	0.031	4	0.040	0.460	0.480	1.4	4.36	300
-33UN	LSFHOF-4	--	14 (class K)	0.031	4	0.040	0.520	0.550	1.6	2.73	250
-34UN	LSFHOF-9	9(90)	--	0.031	4	0.050	0.630	0.660	1.8	1.24	200
-35UN	LSFHOF-42	42 (209)	--	0.070	4	0.070	1.300	1.380	4.1	0.277	150
-36UN	LSFHOF-60	60 (304)	--	0.070	4	0.070	1.430	1.510	4.5	0.191	150
-37UN	LSFHOF-133	133 (684)	--	0.070	4	0.100	1.920	2.000	N/A	0.0847	125

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REQUIREMENTS:

Qualification required.

INSPECTION:

Basic electricals:

Conductor resistance (ohms/1,000 feet at 77 °F [25 °C], max.)	See table I
Voltage withstand (volts, root mean square, min.)	
Type LSSHOF only:	
Conductor to ground	2,000
Types LSDHOF, LSTHOF, LSFHOF only:	
Conductor to conductor (sizes 3 through 9, inclusive)	2,000
Conductor to conductor (sizes 14 through 600, inclusive)	2,500
Insulation resistance (megohms/1,000 feet, min.)	
Type LSSHOF only:	
Conductor to ground table	See table I
Types LSDHOF, LSTHOF, LSFHOF only:	
Conductor to conductor	See table I
Conductor continuity	No failure

Group A:

Visual and dimensional	No failure

Group B:

Thermoset proof test (percent, max.)	
Insulation	50
Jacket (when tested at 392 °F [200 °C])	50
Twisting and bending endurance (see table I for roller size)	
At 167 °F (75 °C) (cycles, min.)	6,000
At -22±3.6 °F (-30±2 °C) (cycles, min.)	2,000

NOTE: Tests to be performed on all sizes of LSDHOF, LSTHOF, and LSFHOF, except LSDHOF-250 and LSDHOF-400, LSTHOF-150 through LSTHOF-600, and LSFHOF-133. Tests are not applicable (N/A) for all sizes of LSSHOF.

Physicals (unaged)

Insulation (extruded insulation only)

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Tensile strength (lb/in^2, min.)	700
Elongation (percent, min.)	150
Jacket (cable)	
Tensile strength (lb/in^2, min.)	1,300
Elongation (percent, min.)	160
Tear (lb/in thickness, min.)	35

Group C:

Physicals (aged) (air oven)	
Insulation (extruded insulation only)	
Tensile strength (percent of unaged, min.)	75
Elongation (percent of unaged, min.)	75
Jacket (cable)	
Tensile strength (percent of unaged, min.)	60
Elongation (percent of unaged, min.)	60
Permanence of printing (insulation, method 1 only) (cycles, min.)	25
Permanence of printing (jacket) (cycles, min.)	125
Heat distortion (percent of unaged, max.)	30
Shrinkage	No failure

Group D:

Flame propagation (cable)	No failure

Qualification inspection:

Qualification inspection shall include basic electricals; groups A, B, C, and D; plus the following:

Aging and compatibility (cable) (257±9 °F [125±5 °C])	No failure
Abrasion resistance (jacket) (scrapes, min.)	75
Acid gas equivalent (percent, max.)	
Jacket	2
Fillers	2
Insulation	18
Halogen content (percent, max.)	
Jacket	0.2
Fillers	0.2
Insulation	0.2

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Immersion (jacket)	
Tensile strength (percent of unaged, min.)	50
Elongation (percent of unaged, min.)	50
Smoke index (max.)	
Jacket	25
Fillers	45
Insulation	45
Toxicity index (max.)	
Jacket	5
Fillers	5
Insulation	1.5
Durometer (jacket) – type A (hardness, min.)	80
Weathering (jacket)	No failure
Electrical moisture absorption	No failure

UNIT ORDERING LENGTHS:

Type	Size	Feet (nominal)
LSSHOF	23	2,000
LSSHOF	60 through 250	1,000
LSSHOF	500 through 800	500
LSDHOF	6	2,000
LSDHOF	9 and 14	1,500
LSDHOF	23 through 83	1,000
LSDHOF	250 and 400	500
LSTHOF	6	2,000
LSTHOF	9 and 14	1,500
LSTHOF	23 through 150	1,000
LSTHOF	250 through 600	500
LSFHOF	4	2,000
LSFHOF	9	1,500
LSFHOF	42 and 60	1,000
LSFHOF	133	500

NOTE: LSSHOF-3, LSDHOF-3, LSDHOF-4, LSTHOF-3, LSTHOF-4, LSFHOF-3 (packed): 500 feet (nominal).

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CHANGES FROM PREVIOUS ISSUE: Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

CONCLUDING MATERIAL

Custodians:
Army – MI
Navy – SH

Preparing activity:
Navy – SH
(Project 6145-2013-020)

Review activities:
Army – AR, CR
Navy – CG, EC
DLA – CC

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at https://assist.dla.mil.

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